PILOT REVIEW · MBA ZG583 MANAGEMENT OF AI PRODUCTS

PolicyGuard AI

A compliance assistant for insurance underwriting document audit and policy renewal

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All data, rules, companies and people are fictional. Business figures are planning assumptions.

PROBLEM AND STAKES

Audit gaps surface too late, when they cost the most

- Every policy folder must meet set rules for structure, naming and content
- Staff check folders one at a time, by hand
- Gaps often surface during a third-party audit, or at renewal
- Extra headcount adds cost, and reviews stay inconsistent

12,000

commercial property folders a year

45 min

analyst time per folder

$1.35M

annual review cost at $150/h onsite

Planning assumptions for a hypothetical mid-size US insurer. Not company data.

The analyst sees a gap first, and answers for it if it is missed

Today: Meera, underwriting operations analyst

Open folder → **Check names** → **Read documents** → **Compare by hand**

Legacy naming varies, so much of the day goes to routine checks.

“I don't want an auditor to find something that I signed off.”

Two decisions the product improves

Each finding
confirm, reject or escalate, with a reason

The renewal
renew, review or hold, on complete evidence

Persona is illustrative and not yet validated through interviews.

Scripts read file names, but this work needs reading documents

Option	Limit
File-name script	Found 8 of 30 planted defects
Repository	Stores documents, and cannot judge them
Checklist	Same manual reading
More analysts	Cost rises with volume

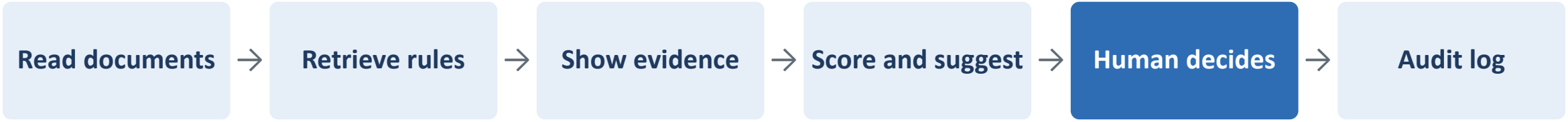
Why now

- Multimodal models read digital and scanned PDFs
- Structured output can require a quote, page and rule for each finding
- APIs remove model training, at an illustrative 2.5 cents a folder

Not automated: sign-off or renewal.

Defect count from synthetic test folders.

One folder, one decision loop, and no chat box



In the MVP

- One commercial property folder at a time
- Structure, naming and content checks
- Rule, quote, page and confidence on each finding
- Renewal suggestion, sign-off and report

Left out on purpose

- Bulk scanning
- Automatic renewal or sign-off
- Pricing and auto-remediation
- Policy system integration

The hard choices: what I rejected, delayed and overrode

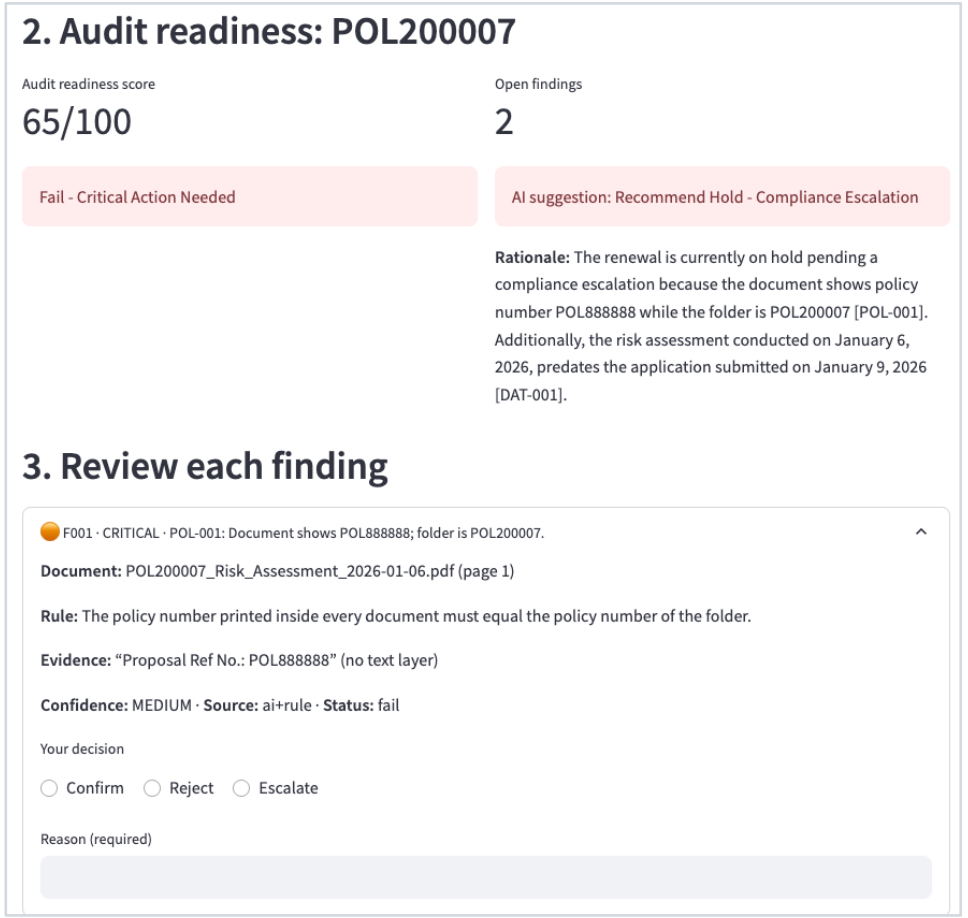
Decision	Call	Reason
AI approves renewals	Rejected	High value on paper, but regulatory and customer harm if wrong
Risk pricing from folders	Rejected	Fairness and regulatory exposure
RAG scored low in RICE	Kept in MVP	It carries the AI hypothesis; without it, this is the same script
Bulk scan	Delayed to Phase 3	An unproven model at scale multiplies its errors
Build vs buy vs API	API-first RAG	Value lies in the insurer's rules and review process, not in owning a model
First wedge	Commercial property	Clear documents, and less personal data than health

Full PRD, RICE and roadmap are in the report and Appendix.

PROTOTYPE WALKTHROUGH

Live demo: a folder with valid names fails on its content

- **Fail, 65/100, Hold:** a wrong policy number and date order, found inside scanned documents
- **Evidence:** rule, quote and why confidence is only MEDIUM
- **Human decision:** a reason is required, and sign-off stays locked until complete
- **Record:** audit log and report
- **AI off:** Not Scored, with no recommendation



Backup screenshot, in case the live demo or network fails. Synthetic folder POL200007.

Testing changed the product, and the results are promising, not proof

12 unseen synthetic folders	PolicyGuard AI	File-name script
Defects found	**16 of 16**	4 of 16
False alarms	1	2
Critical missed (of 3)	0	2
Renewals as expected	12 of 12	n/a

The 95% confidence interval still allows recall near 79%.

What testing exposed

- First prototype scored 100/100 when the AI failed; it now gives Not Scored
- 14 of 14 hid a skipped critical rule; signature rules now always run
- One finding changed between runs; repeatability is now a gate
- Data is tidy and synthetic, so real accuracy is likely to be lower

Holdout set run once, after the code was frozen.

People are the cost, and time savings pay it back

Scenario ($360k one-time)	Net / year	Payback
Downside: time −20%	$164k	26 months
Time savings, −40%	$434k	**10 months**
Time savings, −50%	$569k	**8 months**
−40% + 1 avoided remediation	$584k	7 months
−50% + 1 avoided remediation	$719k	6 months

All people roles at $150/h onsite. Model cost is under 1% of the $178k annual run cost. Avoided remediation is upside with no evidence yet.

Launch blockers

- **Critical recall** below 95% on real data
- **Any fabricated evidence** in the 100-finding audit
- **No privacy approval** for the chosen provider

Metrics to track in the pilot

Analyst time, rejection rate, usefulness, and gaps found that manual review missed

All figures are planning assumptions, to be replaced by Phase 0 measurements.

FINAL RECOMMENDATION

Pilot, not scale

Fund a 12-week controlled pilot: commercial property, 6–8 analysts, shadow mode first.

Enter the pilot if

- Critical recall ≥ 95%
- Fabricated evidence: 0 of 100
- Repeatability ≥ 95%
- Privacy sign-off

Scale only if

- Analyst time −40%
- ≥ 70% rate it useful
- Rejection rate 10–40%
- Funded rulebook owner

Stop if

- Critical misses can't be controlled
- Rejection stays above 40%
- Privacy approval is refused